Introduction

Choosing a college and major is one of the most important decisions students make. This decision is also often influenced by expectations about future salary. Many students assume that attending a prestigious school or choosing a “high-paying” major will guarantee financial success. However, it is unclear which factors truly have the greatest impact on earnings, and whether common assumptions about school prestige and major choice are supported by data.

The “Where It Pays to Attend College” dataset compiles salary data for different colleges and majors. The dataset contains three tables:

<u>Salaries for Colleges by Type:</u> Compares starting and mid-career salaries across different types of schools (party schools, Ivy League, engineering-focused, state schools, etc.)
<u>Salaries for Colleges by Region:</u> Compares salary outcomes across geographic regions (California, Northeast, South, etc.)
<u>Salaries for Majors:</u> Provides salary information across different fields of study (Accounting, Agriculture, Spanish, etc.), including starting salaries and mid-career earnings

The second dataset, “College Majors and Their Graduates,” provides additional information on job outcomes by major. This dataset includes variables such as unemployment rate, median salary, and salary percentiles, allowing us to analyze not only earnings but also job stability and employment opportunities across different fields. The data is taken from a 2025 graduation class, so a recent and reliable dataset.

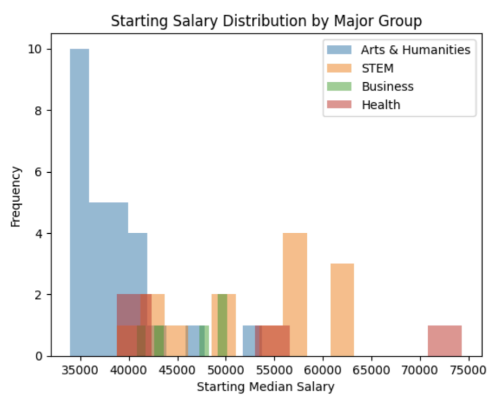
While we do identify what factors lead to higher salaries, our goal is to determine whether these factors are as important as students often believe. Through application of statistical models like linear regression, train test/split, and logistic regression, we aim to provide evidential insights that can help students make more informed decisions about their education and career paths.

We aim to help high school students make a more informed decision of where and what to study and which choices may provide the best return on their educational investment. By analyzing both salary outcomes and job market conditions, students can better understand which factors meaningfully impact earnings and which may be less important than commonly assumed.

The main questions we wish to answer are:

1. Which factor has a greater impact on starting salary: major choice or school type?
2. Which majors are most likely to experience strong salary growth over time?
3. How do job market conditions, such as unemployment rates, relate to salary outcomes across majors?

I. Impact of major group on starting salary

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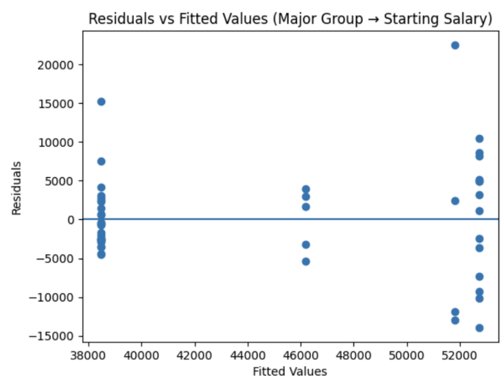
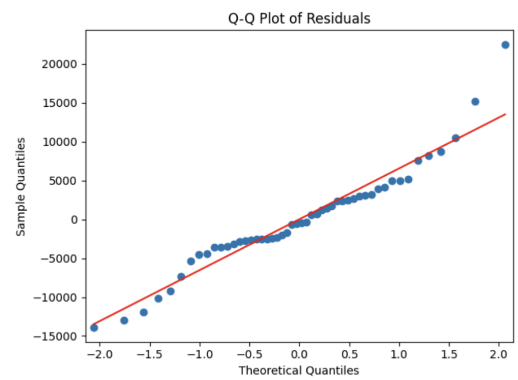
                              OLS Regression Results
==================================================================================
Dep. Variable:     Starting Median Salary   R-squared:                       0.502
Model:                                OLS   Adj. R-squared:                  0.469
Method:                     Least Squares   F-statistic:                     15.43
Date:                    Tue, 21 Apr 2026   Prob (F-statistic):           4.41e-07
Time:                            02:02:20   Log-Likelihood:                -510.25
No. Observations:                      50   AIC:                             1029.
Df Residuals:                          46   BIC:                             1036.
Df Model:                               3
Covariance Type:                nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       3.848e+04   1312.725     29.314      0.000    3.58e+04    4.11e+04
Business    7718.5185   3320.961      2.324      0.025    1033.771    1.44e+04
Health      1.332e+04   3654.472      3.644      0.001    5962.448    2.07e+04
STEM        1.425e+04   2246.476      6.345      0.000    9732.312    1.88e+04
==============================================================================
Omnibus:                        9.130   Durbin-Watson:                   2.106
Prob(Omnibus):                  0.010   Jarque-Bera (JB):               10.773
Skew:                           0.634   Prob(JB):                      0.00458
Kurtosis:                       4.887   Cond. No.                         4.28
==============================================================================
```





A linear regression model was used to examine how major group affects starting salary, with Arts & Humanities as the baseline category. Results indicate that all other major groups earn significantly higher starting salaries than Arts majors. Specifically, STEM majors earn approximately $14,250 more, Health majors earn about $13,320 more, and Business majors earn about $7,718 more on average, with all differences statistically significant. The model explains approximately 50% of the variation in starting salaries ($R^2 = 0.502$), indicating that major choice is a strong determinant of early career earnings.

<u>**Checking assumptions of linear regression**</u>

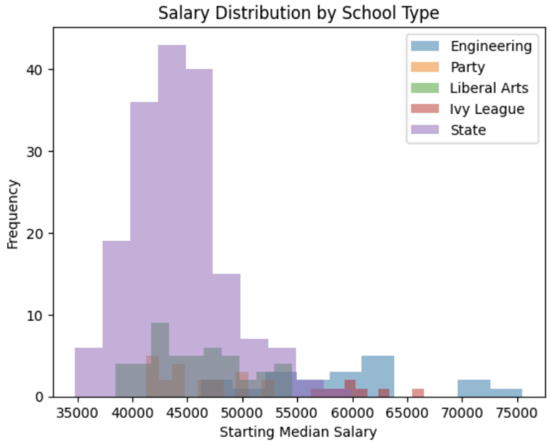
<u>Linearity:</u> The residual plot shows that residuals are generally centered around zero without a clear curve or pattern, suggesting that the linearity assumption is reasonably satisfied. The vertical clustering of points at a few fitted values is expected because the major group is a categorical variable, resulting in only a small number of predicted values.

<u>Constant Variance:</u> The residuals vs. fitted values plot shows that the spread of residuals is not perfectly constant across all fitted values. In particular, there is greater variability for some major groups. This is likely due to the difference in variability across major categories. While the constant variance assumption is not perfectly satisfied, the deviation is not severe enough to invalidate the linear regression results.

<u>Independent errors:</u> Since each observation corresponds to a different majors, there is no evidence of dependence between observations.

<u>Normality:</u> The Q-Q plot shows that the residuals generally follow the reference line, indicating approximate normality. There are slight deviations at the tails, suggesting mild skewness or the presence of outliers, but overall the normality assumption is reasonably satisfied.

II. Impact of school type on starting salary

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                             OLS Regression Results
==============================================================================
Dep. Variable:     Starting Median Salary   R-squared:                       0.502
Model:                                OLS   Adj. R-squared:                  0.495
Method:                     Least Squares   F-statistic:                     66.56
Date:                    Mon, 20 Apr 2026   Prob (F-statistic):           7.06e-39
Time:                            16:41:09   Log-Likelihood:                -2645.5
No. Observations:                     269   AIC:                             5301.
Df Residuals:                         264   BIC:                             5319.
Df Model:                               4
Covariance Type:                nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const         5.906e+04   1045.893     56.466      0.000     5.7e+04    6.11e+04
Ivy League    1417.1053   1921.428      0.738      0.461   -2366.168    5200.379
Liberal Arts -1.331e+04   1239.397    -10.740      0.000   -1.58e+04   -1.09e+04
Party        -1.334e+04   1460.510     -9.136      0.000   -1.62e+04   -1.05e+04
State        -1.493e+04   1101.207    -13.559      0.000   -1.71e+04   -1.28e+04
==============================================================================
Omnibus:                       23.295   Durbin-Watson:                   0.589
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               30.002
Skew:                           0.627   Prob(JB):                     3.06e-07
Kurtosis:                       4.050   Cond. No.                         11.3
==============================================================================
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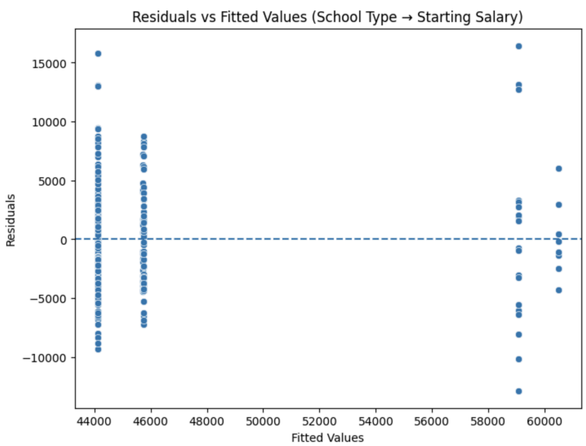

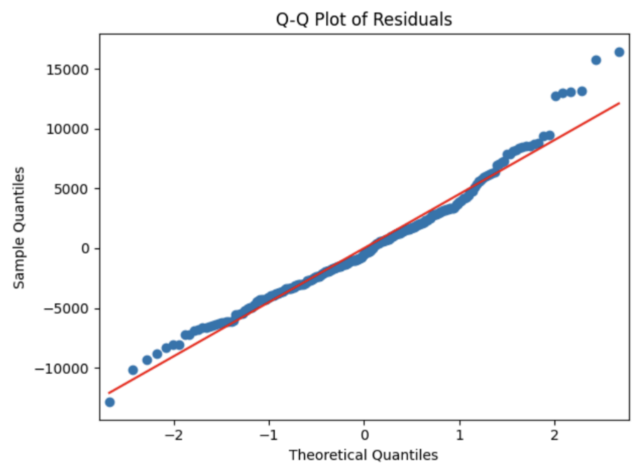

To analyze the impact of school type on starting salary, we used a linear regression model with school type as a categorical variable, using engineering schools as a baseline for comparison. The histogram of salary distributions provides a visual complement to the regression results. It shows that engineering schools tend to have higher starting salaries, with observations concentrated at the upper end of the distribution, while liberal arts, party, and state schools are more concentrated at lower salary ranges. This visual pattern aligns closely with the regression coefficients, which indicate that these school types have significantly lower starting salaries compared to engineering schools.

The regression results indicate that school type explains a substantial portion of variation in starting salaries (R^2 = .502), suggesting that institutional category plays a moderate role in early career earnings. The coefficients show that students coming from liberal arts, party, and state schools earn significantly lower starting salaries compared to engineering schools, with differences ranging from approximately $13,000 to $15,000. These effects are statistically significant, indicating that the observed gaps are unlikely to be due to random chance. In contrast, the coefficient for Ivy League schools is positive but not statistically significant, suggesting that their starting salaries are not meaningfully different from those of engineering schools in this dataset.

In general, these results make sense, however what may make a huge difference for a student selecting a college is whether choosing an Ivy Engineering vs State Engineering school for example as the tuition prices are vastly different different, but have relatively similar salary outcomes.

<u>Checking assumptions of linear regression</u>

<u>Linearity:</u> The residuals are centered around zero without a clear pattern, indicating that a linear relationship is appropriate. The vertical clustering of points at specific fitted values is expected because school type is a categorical variable with only a few distinct groups.

<u>Constant Variance</u>: The spread of residuals across groups look similar, without a strong widening or funnel pattern. This suggests that the model maintains similar prediction accuracy across different school categories.telling us that the assumption of constant variance is reasonably satisfied.

<u>Independent errors</u>: Because each observation represents an individual college and salary outcome rather than repeated measurements over time, the assumption of independent errors is likely satisfied.

<u>Normality:</u> The residuals are not perfectly normally distributed ($p < 0.001$), suggesting that some school types may contain outliers or slightly skewed salary distributions. However, the residuals remain generally centered around zero, and no extreme deviations are visible in the plot. Also, since we use a large dataset, this model is still appropriate for identifying overall trends in how school type relates to starting salary.

III: Is major group or school type better at predicting starting salary

RMSE (Major): 11358.36922622654

RMSE (School Type): 4781.903291777238

A train/test split was used to evaluate predictive performance using root mean squared error. The model using the major group achieved a root mean squared error of approximately $11,358, while the model using school type achieved a much lower RMSE of approximately $4,782. This indicates that school type provides more accurate predictions of starting salary than major groups in this analysis.

Comparing the importance of major and school type

The R^2 values for both models, major group and school type, are nearly identical. This suggests that both what you study and where you study explain a similar portion of the variation in starting salaries. However, since neither model explains all of the variation, it indicates other factors such as individual skills, internships, or job market conditions. When evaluating predictive performance using a train/test split, the model using school type achieved a substantially lower root mean squared error compared to the model using major group. While both factors are similarly useful for explaining variation in salaries, school type provides more accurate predictions of starting salary in this dataset.

One limitation of this analysis is the imbalance in the dataset across school types. There are substantially more observations for party and state schools than for Ivy League schools, which have very few data points. This imbalance may reduce the reliability of estimates for smaller groups and helps explain why some coefficients, such as Ivy League, are not statistically significant despite appearing higher on average. In addition, this analysis considers school type and major group separately, but an ideal model would examine their interaction. Unfortunately, the available datasets do not link individual majors with specific school types, so we are unable to analyze how these factors jointly influence salary outcomes. As a result, the findings should be interpreted as partial effects rather than a complete picture of how education choices impact earnings.

IV: Likelihood of High Salary Growth by Major

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                           Logit Regression Results
==============================================================================
Dep. Variable:            High Growth   No. Observations:                   50
Model:                          Logit   Df Residuals:                       46
Method:                           MLE   Df Model:                            3
Date:                Mon, 20 Apr 2026   Pseudo R-squ.:                  0.2663
Time:                        17:15:47   Log-Likelihood:                -25.430
converged:                      False   LL-Null:                       -34.657
Covariance Type:            nonrobust   LLR p-value:                 0.0003542
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.6931      0.408     -1.698      0.090      -1.493       0.107
Business       2.0794      1.190      1.747      0.081      -0.253       4.412
Health       -16.9376   3368.475     -0.005      0.996   -6619.027    6585.152
STEM           2.4849      0.866      2.869      0.004       0.788       4.182
==============================================================================

Odds Ratios:
 const       5.000000e-01
Business    8.000000e+00
Health      4.406598e-08
STEM        1.200000e+01
```

To examine how salary growth varies by major, we used a logistic regression model where the outcome represented if the major's change from starting to mid- career salary fell into the category of high growth. High salary growth was defined as whether a major's increase from starting to mid-career salary exceeded the median growth across all majors. This creates a binary outcome where approximately half of the majors are classified as high growth. Majors were grouped into four categories: STEM, Business, Health, and Arts & Humanities, with Arts & Humanities serving as the baseline group.

The results show that major category plays a meaningful role in determining the likelihood of high salary growth. In particular, STEM majors have a large positive and statistically significant coefficient ($p = 0.004$), indicating that they are significantly more likely than Arts & Humanities majors to experience high salary growth. The corresponding odds ratio suggests that STEM majors have approximately 12 times higher odds of achieving high growth relative to the baseline group.

Business majors also exhibit a positive coefficient, with an odds ratio of about 8, suggesting substantially higher odds of high salary growth compared to Arts & Humanities. However, this result is only marginally significant ($p = 0.081$), meaning the evidence is weaker. In contrast, Health majors show a large negative coefficient, indicating a much lower likelihood of high salary growth.

This information is important to know before choosing a career because it helps visualize what the future holds in the long run. For example, a less than ideal starting salary/ job is not the end of the world and may just be a part of the process for some careers.

V: How does the job market affect starting salaries across majors?


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                              OLS Regression Results
==================================================================================
Dep. Variable:     Starting Median Salary   R-squared:                       0.280
Model:                                OLS   Adj. R-squared:                  0.248
Method:                     Least Squares   F-statistic:                     8.565
Date:                    Tue, 05 May 2026   Prob (F-statistic):            0.00781
Time:                            14:27:04   Log-Likelihood:                -248.80
No. Observations:                      24   AIC:                             501.6
Df Residuals:                          22   BIC:                             504.0
Df Model:                               1
Covariance Type:                nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const              6.589e+04   6704.821      9.828      0.000      5.2e+04    7.98e+04
Unemployment_rate -2.545e+05    8.7e+04     -2.927      0.008    -4.35e+05   -7.42e+04
==============================================================================
Omnibus:                        4.625   Durbin-Watson:                   2.429
Prob(Omnibus):                  0.099   Jarque-Bera (JB):                1.580
Skew:                           0.048   Prob(JB):                        0.454
Kurtosis:                       1.747   Cond. No.                         53.3
==============================================================================
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To explore how job outlook relates to starting salary, we created a scatterplot comparing unemployment rate to starting salary for each major, with points color-coded by major category. The shaded confidence interval around the regression line reflects uncertainty in the estimated relationship and widens at higher unemployment rates due to fewer observations in that range. Overall, the results suggest that students pursuing majors with stronger job demand may benefit from both higher initial salaries and more stable employment opportunities

A linear regression model was used to quantify this relationship, and a regression line was added to summarize the overall trend. The results reveal a clear negative relationship between unemployment rate and starting salary. Majors with lower unemployment rates tend to have higher starting salaries, while majors with higher unemployment rates are generally associated with lower starting pay. Specifically, the regression coefficient indicates that a one percentage point increase in unemployment rate is associated with an approximate $2,545 decrease in starting salary. The regression results show that unemployment rate is a statistically significant predictor of starting salary, which indicates that job market conditions play an important role in determining early career earnings. Since a large portion of the variation in salary remains unexplained, other conditions such as location or specific jobs within a field play an important role in determining starting salary.

Overall, these findings highlight that majors with lower unemployment rates generally provide both higher salaries and greater job security, making them more attractive options for students seeking a strong balance between earnings and employment stability

Conclusion and recommendations

Overall, the analysis shows that both major choice and school type play important roles in determining starting salary, but neither factor alone fully explains salary outcomes. Major choice emerges as the strongest and most consistent factor. STEM and Health majors tend to earn higher starting salaries, while Arts & Humanities majors consistently earn less. The logistic regression results further show that STEM majors are significantly more likely to experience high salary growth. While some fields offer higher initial pay, others may offer stronger long-term growth opportunities and both should be taken into consideration when choosing a major or career field.

Job market conditions should also be taken into consideration when looking to enter a specific field. Especially in a time with the rising influence of AI affecting many jobs, students should take into account how stable and in demand their field is. Our regression results highlight the importance of not only choosing a major based on interest, but also considering long term employment prospects. Ultimately, students should aim to find a balance between pursuing what they enjoy and selecting a field that offers strong job opportunities and financial stability.

Even though choosing a major, where to attend college, and a potential career path can be very intimidating for many high school students, our analysis helps break down this decision into key factors. The results suggest that students may not need to stress as much about attending a specific type of school as they often believe. Instead, both major choice and school type contribute similarly to salary outcomes, while a large portion of variation is driven by other factors. In particular, major selection plays an important role not only in starting salary but also in long term salary growth. Students may benefit more from focusing on choosing a field that aligns with strong career opportunities, job market conditions and personal interests rather than relying solely on school prestige.

References

Datasets:

Where it Pays to Attend College:
https://www.kaggle.com/datasets/wsj/college-salaries?select=degrees-that-pay-back.csv

College Majors and their Graduates:
https://www.kaggle.com/datasets/thedevastator/uncovering-insights-to-college-majors-and-their?resource=download

Code: https://colab.research.google.com/drive/1mqZuce-N3juJWrAadym41wkpzprS1luP?usp=sharing